



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 27/02/2020*

EXERCISE

We observe N samples (for $n=0, 1, \dots, N-1$) of a realization of the random process $Z[n] = \sqrt{\tau} \cdot X[n]$, where τ is the so-called signal “texture”, modelled as an unknown deterministic parameter, and $X[n]$ is multiplicative noise, typically called “speckle”, modelled a zero-mean AR(1) Gaussian process with one-lag correlation coefficient ρ ($0 \leq |\rho| < 1$) and power $\sigma_x^2 = 1$.

(1) Derive the Autocovariance Function of the observed process $Z[n]$. Then, derive the log-likelihood function $LLF(\tau) = \ln f_z(\mathbf{z}; \tau)$, where $f_z(\mathbf{z}; \tau)$ is the probability density function (PDF) of the observed data vector $\mathbf{Z} = [Z[0] \ Z[1] \ \dots \ Z[N-1]]^T$.

(2) Derive the Method of Moments (MM) estimator of the texture τ , calculate its bias and its Mean Square Error (MSE); show that the MSE can be expressed as a function of $\alpha(\rho)$, where

$$\alpha(\rho) \triangleq \sum_{m=-(N-1)}^{N-1} \left(1 - \frac{|m|}{N}\right) \rho^{2|m|}. \text{ Establish if the MM estimator is consistent or not.}$$

(3) Derive the Maximum Likelihood (ML) estimator of the signal texture τ , calculate its bias and its Mean Square Error (MSE). Establish if ML estimator is consistent or not.

(4) Find the Cramér-Rao bounds (CRBs) on the estimation of τ through the definition of $CRB(\tau)$.

(5) Compare the MM estimator with the ML estimator when $\rho = 0$.

Useful relationships:

For a zero mean Gaussian process $Z[n]$, we have:

$$E\{Z[n]Z[i]Z[k]Z[l]\} = E\{Z[n]Z[i]\}E\{Z[k]Z[l]\} + E\{Z[n]Z[k]\}E\{Z[i]Z[l]\} + E\{Z[n]Z[l]\}E\{Z[i]Z[k]\}$$

$\mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = \text{Tr}\{\mathbf{R}^{-1} \mathbf{z} \mathbf{z}^T\}$, where $\text{Tr}\{\cdot\}$ is the trace of a square matrix.

Double Sum formula :
$$\sum_{n=0}^{N-1} \sum_{i=0}^{N-1} f[n-i] = \sum_{m=-(N-1)}^{N-1} (N - |m|) f[m]$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

Solution.

(1) Derive the Autocovariance Function of the observed process $Z[n]$. Then, derive the log-likelihood function $LLF(\tau) = \ln f_Z(\mathbf{z}; \tau)$.

$$E\{Z[n]\} = E\{\sqrt{\tau} \cdot X[n]\} = \sqrt{\tau} E\{X[n]\} = 0$$

$$C_Z[m] = R_Z[m] = E\{Z[n]Z[n+m]\} = \tau E\{X[n]X[n+m]\} = \tau R_X[m] = \tau \rho^{|m|}$$

$$\sigma_Z^2 = C_Z[0] = R_Z[0] = \tau$$

$$\mathbf{z} \in N(\mathbf{0}, \mathbf{C}_Z), \quad \mathbf{C}_Z = \tau \mathbf{R}, \quad \text{where } [\mathbf{R}]_{i,j} = \rho^{|i-j|}, i, j = 1, 2, \dots, N$$

$$f_Z(\mathbf{z}; \tau) = \frac{1}{(2\pi)^{N/2} \sqrt{|\tau \mathbf{R}|}} \exp\left(-\frac{1}{2} \mathbf{z}^T (\tau \mathbf{R})^{-1} \mathbf{z}\right) = \frac{1}{(2\pi)^{N/2} \sqrt{\tau^N |\mathbf{R}|}} \exp\left(-\frac{1}{2\tau} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}\right)$$

$$LLF(\tau) = \ln f_Z(\mathbf{z}; \tau) = \ln \left[(2\pi)^{-N/2} |\mathbf{R}|^{-1/2} \right] - \frac{N}{2} \ln \tau - \frac{1}{2\tau} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = k - \frac{N}{2} \ln \tau - \frac{1}{2\tau} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}$$

(2) Derive the Method of Moments (MM) estimator of the signal texture τ , its bias and its Mean Square Error (MSE).

$$\tau = R_Z[0] = E\{Z^2[n]\} \rightarrow \hat{\tau}_{MM} = \frac{1}{N} \sum_{n=0}^{N-1} Z^2[n] = \frac{1}{N} \mathbf{z}^T \mathbf{z}$$

$$E\{\hat{\tau}_{MM}\} = \frac{1}{N} \sum_{n=0}^{N-1} E\{Z^2[n]\} = \frac{1}{N} \sum_{n=0}^{N-1} R_Z[0] = \frac{1}{N} \sum_{n=0}^{N-1} \tau = \tau, \text{ the MM estimator is unbiased.}$$

$$E\{\hat{\tau}_{MM}^2\} = E\left\{\frac{1}{N} \sum_{n=0}^{N-1} Z^2[n] \frac{1}{N} \sum_{i=0}^{N-1} Z^2[i]\right\} = \frac{1}{N^2} \sum_{n=0}^{N-1} \sum_{i=0}^{N-1} E\{Z^2[n]Z^2[i]\}$$

$$\begin{aligned} E\{Z^2[n]Z^2[i]\} &= E\{Z[n]Z[n]Z[i]Z[i]\} = E\{Z^2[n]\}E\{Z^2[i]\} + 2E\{Z[n]Z[i]\}E\{Z[n]Z[i]\} \\ &= R_Z^2[0] + 2R_Z^2[n-i] = \tau^2 + 2(\tau \rho^{|n-i|})^2 = \tau^2 (1 + 2\rho^{2|n-i|}) \end{aligned}$$

$$E\{\hat{\tau}_{MM}^2\} = \frac{1}{N^2} \sum_{n=0}^{N-1} \sum_{i=0}^{N-1} \tau^2 (1 + 2\rho^{2|n-i|}) = \tau^2 + \frac{2\tau^2}{N^2} \sum_{n=0}^{N-1} \sum_{i=0}^{N-1} \rho^{2|n-i|}$$

$$MSE\{\hat{\tau}_{MM}\} = \text{var}\{\hat{\tau}_{MM}\} = E\{\hat{\tau}_{MM}^2\} - (E\{\hat{\tau}_{MM}\})^2 = \frac{2\tau^2}{N^2} \sum_{n=0}^{N-1} \sum_{i=0}^{N-1} \rho^{2|n-i|} = \frac{2\tau^2}{N} \sum_{m=-(N-1)}^{N-1} \left(1 - \frac{|m|}{N}\right) \rho^{2|m|}$$

$$MSE\{\hat{\tau}_{MM}\} = \frac{2\tau^2}{N} \alpha(\rho), \quad \text{where } \alpha(\rho) \triangleq \sum_{m=-(N-1)}^{N-1} \left(1 - \frac{|m|}{N}\right) \rho^{2|m|}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$MSE\{\hat{\tau}_{MM}\} = \frac{2\tau^2}{N} \sum_{m=-(N-1)}^{N-1} \left(1 - \frac{|m|}{N}\right) \rho^{2|m|} \leq \frac{2\tau^2}{N} \sum_{m=-(N-1)}^{N-1} \rho^{2|m|} = \frac{2\tau^2}{N} \left(2 \sum_{m=0}^{N-1} \rho^{2m} - 1\right) = \frac{2\tau^2}{N} \left(2 \cdot \frac{1-\rho^{2N}}{1-\rho} - 1\right)$$

$$\lim_{N \rightarrow \infty} MSE\{\hat{\tau}_{MM}\} \leq \lim_{N \rightarrow \infty} \frac{2\tau^2}{N} \left(2 \cdot \frac{1-\rho^{2N}}{1-\rho} - 1\right) = 0 \Rightarrow \lim_{N \rightarrow \infty} MSE\{\hat{\tau}_{MM}\} = 0$$

Hence, the MM estimator is consistent.

(3) Derive the Maximum Likelihood (ML) estimator of the signal texture τ , its bias and its Mean Square Error (MSE).

$$LLF(\tau) = \ln f_Z(\mathbf{z}; \tau) = \ln \left[(2\pi)^{-N/2} |\mathbf{R}|^{-1/2} \right] - \frac{N}{2} \ln \tau - \frac{1}{2\tau} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = k - \frac{N}{2} \ln \tau - \frac{1}{2\tau} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}$$

$$\frac{dLLF(\tau)}{d\tau} = -\frac{N}{2\tau} + \frac{1}{2\tau^2} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = 0 \rightarrow \hat{\tau}_{ML} = \frac{1}{N} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}$$

$$E\{\hat{\tau}_{ML}\} = \frac{1}{N} E\{\mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}\} = \frac{1}{N} E\{Tr\{\mathbf{R}^{-1} \mathbf{z} \mathbf{z}^T\}\} = \frac{1}{N} Tr\{\mathbf{R}^{-1} E\{\mathbf{z} \mathbf{z}^T\}\} = \frac{1}{N} Tr\{\mathbf{R}^{-1} \tau \mathbf{R}\} = \frac{\tau}{N} Tr\{\mathbf{I}\} = \tau$$

The ML estimator is unbiased. We can observe that:

$$\frac{dLLF(\tau)}{d\tau} = -\frac{N}{2\tau} + \frac{1}{2\tau^2} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = \frac{N}{2\tau^2} \left(\frac{1}{N} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} - \tau \right),$$

so we can say that $\hat{\tau}_{ML} = \frac{1}{N} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}$ is an unbiased and efficient estimator of τ and the CRB is given

by the reciprocal of the Fisher Information: $CRB(\tau) = \frac{1}{N/2\tau^2} = \frac{2\tau^2}{N} = MSE\{\hat{\tau}_{ML}\}$. The ML

estimator is consistent.

(4) Find the Cramér-Rao bounds (CRBs) on the estimation of τ .



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$\frac{d^2 \ln f_z(\mathbf{z}; \tau)}{d\tau^2} = \frac{d}{d\tau} \left(-\frac{N}{2\tau} + \frac{1}{2\tau^2} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} \right) = \frac{N}{2\tau^2} - \frac{1}{\tau^3} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z}$$

$$-E \left\{ \frac{d^2 \ln f_z(\mathbf{z}; \tau)}{d\tau^2} \right\} = -\frac{N}{2\tau^2} + \frac{1}{\tau^3} E \left\{ \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} \right\} = -\frac{N}{2\tau^2} + \frac{N}{\tau^2} = \frac{N}{2\tau^2}$$

$$CRB(\tau) = \frac{1}{-E \left\{ \frac{d^2 \ln f_z(\mathbf{z}; \tau)}{d\tau^2} \right\}} = \frac{2\tau^2}{N}$$

5) Compare the two estimators when $\rho = 0$.

$$\text{If } \rho = 0 \text{ then } \mathbf{R} = \mathbf{I} \rightarrow \hat{\tau}_{ML} = \frac{1}{N} \mathbf{z}^T \mathbf{R}^{-1} \mathbf{z} = \frac{1}{N} \mathbf{z}^T \mathbf{z} = \hat{\tau}_{MM}, \quad MSE \{ \hat{\tau}_{ML} \} = MSE \{ \hat{\tau}_{MM} \} = \frac{2\tau^2}{N}.$$

The two estimators coincide.